

Khushneet Singh [ksploitx.dev](https://github.com/ksploitx)

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EDUCATION

B.Tech, CSE Cyber Security

2022 - 2026

Lakshmi Narain College of Technology and Science, Bhopal, M.P

ACHIEVEMENTS & RESEARCH

AMD SlingShot Finalist

 [Autism Care Ecosystem](#)

- Selected among top teams nationally for ACE Mobile, an AI-powered Flutter app for autism care.
- M-CHAT, TFLite assessments & LLM summaries power continuous autism management.

SIH Grand Finalist

 [Post](#)

- Spearheaded the design and hardware integration of an interactive kiosk system for tourist assistance, built using Electron and React for a responsive, information-rich interface.
- Engineered real-time content synchronization across distributed Raspberry Pi units using MQTT and REST APIs, ensuring consistent, low-latency updates network-wide.

Research Paper: AI-Enhanced Gunshot Detection System

 [View Paper](#)

Co-authored a paper on FPGA-based AI gunshot detection with LoRaWAN integration for border surveillance, covering direction-of-arrival estimation and real-time acoustic event analysis.

PROJECTS

Self-Improving AI Pipeline

 [Weave](#)

- Built a multi-agent LLM orchestration system using LangGraph and FastAPI with dynamic routing, per-agent token budgets, and real-time SSE streaming across 6 specialised AI agents
- Implemented a self-improving eval pipeline with 15 test cases scored across 6 dimensions; a Meta-agent auto-proposes prompt rewrites from failures under human-in-the-loop review
- Designed the backend infrastructure with PostgreSQL, Redis/Celery, FAISS vector search, Alembic migrations, and Docker Compose for multi-service deployment

Workbench (RBAC Admin Panel)

 [WorkBench-RBAC](#)

- Built a role-based access control admin panel in Next.js 14 with system and custom roles
- Designed granular, string-based permission system supporting multi-role assignment per user with union-based effective permissions
- Implemented safeguards preventing system role deletion and automatic cleanup of orphaned role references

Support Ticket Classifier

 [Ticket-Classifier](#)

- Built a hybrid rule-based + ML ticket classification pipeline using TF-IDF and Calibrated LinearSVC (scikit-learn)
- Designed confidence-based routing with optional LLM fallback for ambiguous cases

Dart Multilingual Dependency

 [AutoLingo](#)

- Published a Flutter package on pub.dev that automates multilingual support using Lingo.dev
- Built runtime/build-time UI text detection and translation, removing ARB files and manual localization keys
- Reduced localization setup overhead for Flutter developers building global-ready apps

SKILLS

Programming Languages	Python, JavaScript ,TypeScript, Dart, PostgreSQL, pgvector
Frameworks & Libraries	FastAPI, Django REST Framework(DRF), React, Node.js, Next.js, Flutter, LangChain, LangGraph, Google ADK, FAISS
Platform & DevTools	Git, Docker, Postman, Github Actions, Semantic Search(Exa), Raspberry Pi, REST APIs, FireBase, SupaBase